Backhaus Kaya, Part 4: Eighteen hundred, every night

Case study · Sustainable Supply Chain Management · Session 5 · Forecasting (Chapter 4)

Act 1 · The proposal

Monday, 07:15. The complaint ledger is history. Four weeks after Tarek re-set the divider, FrischMarkt's stores have logged nine complaints instead of a hundred and thirty, and Frau Lindner's e-mail this morning begins with a word Aylin Kaya has never read from her before: *Congratulations.* Then comes the second paragraph.

> **From:** M. Lindner, FrischMarkt purchasing · **Subject:** Next step – quantities per store
>
> Dear Ms Kaya, the rolls are right. The quantities are not. We take 1,800 rolls every night, split 720 · 660 · 420 across our three stores, and every morning two things happen at once: the Altstadt store throws rolls away on Monday, and the Campus store is sold out by ten on Saturday. We propose to order **a quantity per store and per night** from next month. To help you, I attach twelve weeks of our till data for all three stores. Please send your proposal for week 13 by Friday. – M. Lindner

Forecast = a statement about the future made from data – here: how many rolls each store will sell tomorrow night. **Till data** = what the shop's cash register recorded: rolls sold, night by night. **Returns** = rolls delivered but not sold; the store sends them back and Kaya throws them away. **Demand** = what customers wanted to buy, whether or not the shelf still had it.

Aylin opens the attachment: three columns of numbers, seventy-two nights each (Exhibit A). Last month's carbon plan is pinned to the wall behind her, and on it is one line she has never liked: *6 % of the rolls come back – 32,400 rolls a year baked for the bin.* At 0.10 € of flour, power and bags each, and 56.5 g CO_2e each.

At 08:00 Jonas Weber reads the e-mail standing up. "Eighteen hundred, every night. We have done that for three years. The stores sell what we bring; if they don't, that is their shop window, not my oven." He taps the Campus column. "And this one wants more? Then she should order more."

Lena, Aylin's niece, is home for the semester break and has already pasted the three columns into a spreadsheet. "They don't sell what we bring. Look at Mondays. Every Monday in Altstadt is under six hundred, every Saturday is over eight hundred, and we deliver 720 on both. And Campus is not the same shop it was in week one." Jonas: "Students. Who can forecast students?" Lena: "Anyone with a calendar."

Aylin looks at the sign over the oven. *Never promise bread you cannot bake.* She takes a marker and adds a line underneath: *and never bake bread nobody will buy.* Frau Lindner wants a proposal by Friday. Aylin wants to know, before she writes a single number, whether a forecast can be trusted more than Jonas's three years of feel.

Your prediction (team, 5 minutes, no calculator). Write each answer on its own sticky note with your team code name. Professor Bartnik collects them – we check them at noon.

P1 Which weekday needs the most rolls in Altstadt – and how many more than a Monday, in %?

P2 Of the 32,400 rolls returned each year, how many can a per-store forecast save: a quarter, half, two thirds?

P3 Which store has changed over the twelve weeks, and how?

Exhibit A · Rolls demanded per store and night, 12 weeks (72 delivery nights, FrischMarkt till data)

Night	Wk	Day	Altstadt	Bahnhof	Campus
1	1	Mon	553	519	343
2	1	Tue	626	587	391
3	1	Wed	652	600	375
4	1	Thu	689	647	393
5	1	Fri	704	679	417
6	1	Sat	868	733	484
7	2	Mon	580	528	357
8	2	Tue	673	536	356
9	2	Wed	608	584	414
10	2	Thu	683	587	408
11	2	Fri	736	639	430
12	2	Sat	823	724	484
13	3	Mon	596	524	339
14	3	Tue	631	609	391
15	3	Wed	688	572	392
16	3	Thu	685	598	429
17	3	Fri	652	674	438
18	3	Sat	812	755	509
19	4	Mon	597	518	369
20	4	Tue	652	594	419
21	4	Wed	683	603	422
22	4	Thu	699	642	434
23	4	Fri	700	662	425
24	4	Sat	789	717	509
25	5	Mon	578	517	373
26	5	Tue	611	586	416
27	5	Wed	642	597	407
28	5	Thu	704	574	422
29	5	Fri	656	619	441
30	5	Sat	791	763	533
31	6	Mon	579	520	359
32	6	Tue	669	584	400
33	6	Wed	649	621	417
34	6	Thu	685	610	445
35	6	Fri	708	630	475
36	6	Sat	832	745	521

Night	Wk	Day	Altstadt	Bahnhof	Campus
37	7	Mon	555	510	341
38	7	Tue	629	593	398
39	7	Wed	648	631	424
40	7	Thu	642	**752***	452
41	7	Fri	711	**854***	444
42	7	Sat	845	**997***	548
43	8	Mon	564	536	368
44	8	Tue	654	583	414
45	8	Wed	697	580	427
46	8	Thu	684	601	446
47	8	Fri	723	625	405
48	8	Sat	788	770	545
49	9	Mon	564	510	388
50	9	Tue	659	607	415
51	9	Wed	662	602	451
52	9	Thu	665	588	431
53	9	Fri	687	631	460
54	9	Sat	797	741	568
55	10	Mon	573	532	378
56	10	Tue	670	588	434
57	10	Wed	658	590	427
58	10	Thu	693	591	441
59	10	Fri	711	600	476
60	10	Sat	821	703	548
61	11	Mon	574	513	403
62	11	Tue	636	574	419
63	11	Wed	618	580	459
64	11	Thu	669	596	491
65	11	Fri	683	618	513
66	11	Sat	821	777	533
67	12	Mon	607	564	399
68	12	Tue	650	586	449
69	12	Wed	689	619	454
70	12	Thu	725	615	447
71	12	Fri	676	623	469
72	12	Sat	817	702	543

FrischMarkt till data; on the few mornings a shelf was empty the staff counted the customers who asked, so the table is **demand, not sales**. Jonas's split today: 720 · 660 · 420 = 1,800 rolls, every night. Shaded: Altstadt, weeks 7–12 – the nights your team calculates with. * = Bahnhof, week 7, city festival (Stadtfest), Thu-Sat.

Act 2 · Lena's three methods

On Tuesday evening Lena has three sheets of paper on the kitchen table, one per store, and a question for Aylin: "Before we send FrischMarkt a number, we need to know how wrong we will be. Every forecast is wrong. The question is by how much, and whether the miss is a pattern we could have used."

> **Time series** = the same quantity measured at regular times – here one number per store per night. Its parts: a **trend** (a long slope up or down), **seasonality** (a pattern that repeats with a fixed length – here the six weekdays), and **random variation** (the noise that is left when trend and pattern are taken out). **Error** = actual − forecast. **MAD** (mean absolute deviation) = the average size of the miss, in rolls. **MAPE** (mean absolute percent error) = the average miss as a percentage of what really sold.

Lena starts with the simplest shop, Altstadt, and the simplest methods: forecast tonight from last night; from the average of the last three nights; from a running average that trusts the

past more or less (**exponential smoothing**, with a smoothing constant α between 0 and 1: near 0 = calm, near 1 = nervous). Then she tries what Jonas said nobody can do: she gives every weekday its own factor. Finally she holds Jonas's Campus rule – 420 rolls a night, every night – against twelve weeks of reality, the way Lena held the divider against the control chart last month.

Aylin has until Friday. She needs: how big the miss of each method is on the last two weeks, which method to trust, what the weekday pattern is, whether the Campus rule is still right, and what the whole exercise is worth in rolls, euros and carbon. **Your team is Lena's team.** Split the work: one pair takes Q1, one pair Q2, then everybody Q3–Q5.

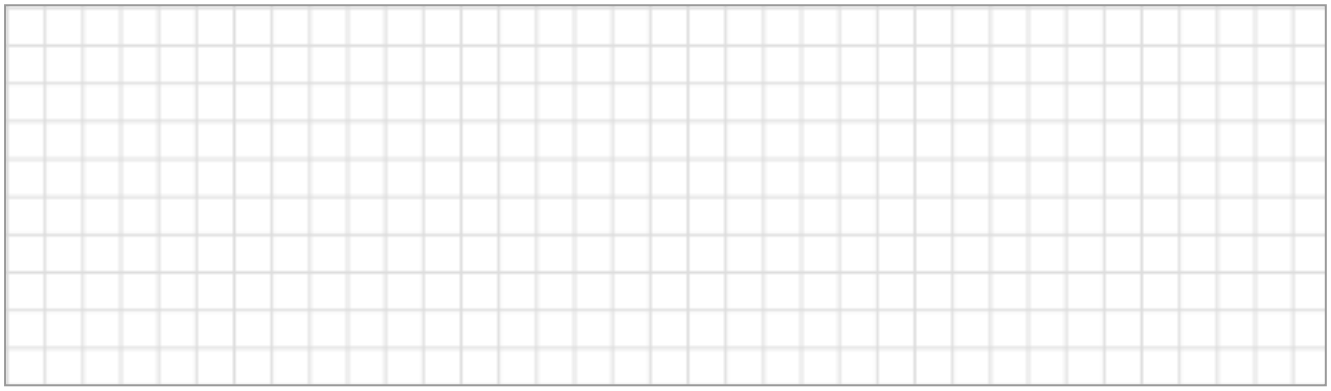
Questions for your team

Q1 · Naive and moving average. Use Exhibit A, Altstadt, nights 61–72 (weeks 11–12). The history you need: night 58 = 693, night 59 = 711, night 60 = 821. **Naive: forecast for tonight = last night's actual. Moving average (3 nights): forecast = (the last three actuals added up) ÷ 3, rounded to the whole roll.** For every night write the forecast and **|error| = the miss without its sign**. Then **MAD = sum of |error| ÷ number of nights** (two decimals). Which of the two misses less?

Q2 · Exponential smoothing, calm and nervous. Same nights. **New forecast = old forecast + α × (last actual − old forecast), rounded to the whole roll before the next step; x.5 rounds up.** Start value for night 61: the week-10 average, **688** rolls. Run it twice: α = 0.2 and α = 0.5. MAD of each. Which α fits Altstadt better, and what does the other one do on the Saturday jump?

Night	Day	Actual	Naive fc	|e|	MA3 fc	|e|	ES α 0.2 fc	|e|	ES α 0.5 fc	|e|
61	Mon	**574**								
62	Tue	**636**								
63	Wed	**618**								
64	Thu	**669**								
65	Fri	**683**								
66	Sat	**821**								
67	Mon	**607**								
68	Tue	**650**								
69	Wed	**689**								
70	Thu	**725**								
71	Fri	**676**								
72	Sat	**817**								
Sum of |e| · MAD = sum ÷ 12										

Chart: plot the twelve actuals (dots) and one forecast (line) – the naive line is the actual line shifted one night to the right. Nights 61–72 from left to right; 5 mm = 20 rolls, bottom line = 540.



Q3 · The weekday pattern. Use Exhibit A, Altstadt, weeks 7–10 (nights 37–60, the first shaded block). Write the four values of every weekday into the grid. **Seasonal (weekday) index = average of that weekday ÷ average of all nights** (two decimals); the six indices add up to 6.00. Then the proposal for Frau Lindner: **seasonal forecast = level × index, where level = last week's average (whole rolls), rounded to the whole roll.** Week 12 in Altstadt was Mon 607 · Tue 650 · Wed 689 · Thu 725 · Fri 676 · Sat 817. Forecast the six nights of week 13 and add them up. Compare with Jonas's 720 × 6 = 4,320.

Weekday	Week 7	Week 8	Week 9	Week 10	Sum	Average (sum ÷ 4)	Index = avg ÷ overall
Mon							
Tue							
Wed							
Thu							
Fri							
Sat							
All 24 nights: sum _____ · overall average = sum ÷ 24 = _____							**Σ index = _____**

Week 13 forecast	Mon	Tue	Wed	Thu	Fri	Sat	Week total
level × index =							

Q4 · Which method wins? Back to nights 61–72. First the fifth method: the seasonal forecast for each night = **level × index**, where level = the average of the week before (week 10 = 688 for nights 61–66; week 11 = the average of nights 61–66, rounded, for nights 67–72), with your indices from Q3. Write its twelve forecasts and |errors|. Then for all five methods: **MAPE = sum of (|error| ÷ actual × 100 %) ÷ number of nights** (two decimals). Which method wins, and by how much? Is the ranking by MAPE the same as by MAD?

Night	61 Mon	62 Tue	63 Wed	64 Thu	65 Fri	66 Sat	67 Mon	68 Tue	69 Wed	70 Thu	71 Fri	72 Sat
Seasonal fc												
|e|												

Method (nights 61–72)	Sum of |e|	MAD	Sum of |e| ÷ actual × 100 %	MAPE
Naive				
3-night moving average				
Exponential smoothing α = 0.2				
Exponential smoothing α = 0.5				
Weekday index × last week's average				

Q5 · Is Jonas's Campus rule still right? Exhibit A, Campus, all twelve weeks, added up per week (the actual column is filled in for you). Jonas's rule: 420 rolls a night = **2,520 a week**, every week. **Tracking signal = RSFE ÷ MAD, where RSFE = all errors so far added up (with signs) and MAD = all |errors| so far ÷ number of weeks so far** (one decimal). The limits are **±4**. In which week does the signal first leave the limits? Is the rule noise, or biased – and in which direction? What happened at Campus?

Week	Actual (Campus, 6 nights)	Forecast	Error = actual − forecast	RSFE (errors added up)	Σ |e| so far	MAD = Σ|e| ÷ weeks	TS = RSFE ÷ MAD
1	**2,403**	2,520					
2	**2,449**	2,520					
3	**2,498**	2,520					
4	**2,578**	2,520					
5	**2,592**	2,520					
6	**2,617**	2,520					
7	**2,607**	2,520					
8	**2,605**	2,520					
9	**2,713**	2,520					
10	**2,704**	2,520					
11	**2,818**	2,520					
12	**2,761**	2,520					

Q6 · Three sentences for Aylin. (a) The proposal for Frau Lindner: the six weekday quantities for Altstadt in week 13 (from Q3), and one sentence on why they beat "720 every night". (b) What the trial says about the other two stores – the Bahnhof festival week (which kind of knowledge forecasts that: the data or the calendar?) and Campus (what to change in the rule). (c) If the per-store forecast cuts returns from 6 % to 2 % of 540,000 rolls: how many rolls a year are no longer baked for the bin, what are they worth at 0.10 € each, and how much CO_2e at 56.5 g per roll (Week 3)? Compare with the best card of the Week 3 game (M1, 6.86 t for 662 € a year).

Backhaus Kaya, FrischMarkt and all persons are fictional; the numbers are teaching values. Textbook: Heizer, J., Render, B., & Munson, C. (2023). *Operations management: Sustainability and supply chain management* (14th ed., Global ed.). Pearson. Chapter 4. Written with Claude (Anthropic); all figures computed in Python; reviewed by and responsibility of Roman Bartnik.